

iClusterBayes Summary Report

Automated Report

Hyperparameters

```
cat('sdev = 0.005, beta.var.scale = 0.3\n')
```

```
## sdev = 0.005, beta.var.scale = 0.3
```

Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.6333  0.8428  0.8494  0.8467  0.8567  0.8756

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09111 0.13778 0.13312 0.16944 0.87722
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.09444 0.14111 0.13591 0.17222 0.74500
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.005000 0.008333 0.100723 0.030556 1.000000
```

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Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0005556 0.0050000 0.0083333 0.0766210 0.0233333 1.0000000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0061111 0.015556 0.164932 0.123889 1.000000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.4944  0.6067  0.6572  0.6525  0.7022  0.8006

beta.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04500 0.08889 0.09685 0.13167 0.83222
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1      1      1        1      1      1

gamma.ar:
Modality 1 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.04667 0.09222 0.09896 0.13556 0.61611
Modality 2 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0094444 0.1635265 0.3694444 1.0000000
Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.875   1.000   1.000   0.995   1.000   1.000
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0016667 0.0932908 0.0647222 0.9633333
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.03111 0.36500 0.31859 0.50722 1.00000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.9428  0.9928  0.9950  0.9944  0.9967  1.0000

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02722 0.06111 0.07349 0.10000 0.84667
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.02889 0.06333 0.07567 0.10333 0.68333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001263 0.0000000 0.5505556
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001169 0.0000000 0.0588889
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0003355 0.0000000 0.3711111
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0000000 0.0000000 0.0001083 0.0000000 0.2022222

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.9378 0.9917 0.9939 0.9934 0.9956 1.0000

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01722 0.04389 0.06147 0.08111 0.84611
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01778	0.04556	0.06279	0.08444	0.71056

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.688e-05	0.000e+00	1.439e-01

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	4.741e-06	0.000e+00	7.778e-03

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	5.669e-06	0.000e+00	1.667e-03

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00e+00	0.00e+00	0.00e+00	8.84e-05	0.00e+00	1.00e+00

--- Summary for K = 5 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9578	0.9922	0.9939	0.9938	0.9956	0.9994

beta.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01222	0.03056	0.04681	0.06167	0.89167

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.01333	0.03333	0.04830	0.06556	0.61778

Modality 2 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	9.990e-07	0.000e+00	1.667e-02

Modality 3 :					
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	1.549e-07	0.000e+00	1.667e-03

Modality 4 :					
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Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.543e-06	0.000e+00	5.556e-03

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	0.000e+00	3.020e-07	0.000e+00	4.444e-03

--- Summary for K = 6 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.5094	0.7000	0.7511	0.7423	0.7922	0.8822

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008333	0.022222	0.035896	0.045556	0.855556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.008889	0.023333	0.036711	0.048889	0.502778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0009271	0.0000000	0.7466667

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000	0.3744	0.6094	0.5584	0.7722	1.0000

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0000000	0.0000000	0.0000000	0.0002462	0.0000000	0.1877778

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.000000	0.000000	0.001878	0.000000	1.000000

--- Summary for K = 7 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9683	0.9917	0.9933	0.9934	0.9950	0.9994

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
------	---------	--------	------	---------	------

0.000000 0.007222 0.017222 0.030347 0.037222 0.857778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

gamma.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.007778	0.018889	0.031161	0.040000	0.487778

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	0	0	0	0	0

--- Summary for K = 8 ---

Z.ar:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9739	0.9917	0.9933	0.9931	0.9950	0.9989

beta.ar:

Modality 1 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000000	0.005556	0.013889	0.025893	0.031667	0.745556

Modality 2 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 3 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 4 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

Modality 5 :

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1	1	1	1	1	1

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.006667 0.014444 0.026777 0.033333 0.445000
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 3.950e-06 0.000e+00 2.239e-01
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000e+00 0.000e+00 0.000e+00 9.106e-05 0.000e+00 1.428e-01
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 2.72e-07 0.00e+00 1.00e-02

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
 0.9556  0.9906  0.9928  0.9923  0.9944  0.9994

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005000 0.01278 0.02645 0.02833 0.82167
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.000000 0.005556 0.013333 0.026209 0.031111 0.648889
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
0.00e+00 0.00e+00 0.00e+00 3.53e-05 0.00e+00 1.00e+00
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    0         0         0         0         0         0
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
    0         0         0         0         0         0
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.    Max.

```

```
0.0000000 0.0000000 0.0000000 0.0000247 0.0000000 0.5366667
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6333 0.8428 0.8494 0.8467 0.8567 0.8756
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09111 0.13778 0.13312 0.16944 0.87722
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.09444 0.14111 0.13591 0.17222 0.74500
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005000 0.008333 0.100723 0.030556 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0005556 0.0050000 0.0083333 0.0766210 0.0233333 1.0000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006111 0.015556 0.164932 0.123889 1.000000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4944 0.6067 0.6572 0.6525 0.7022 0.8006
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.00000 0.04500 0.08889 0.09685 0.13167 0.83222
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.04667 0.09222 0.09896 0.13556 0.61611
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0094444 0.1635265 0.3694444 1.0000000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.875 1.000 1.000 0.995 1.000 1.000
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000000 0.0005556 0.0016667 0.0932908 0.0647222 0.9633333
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.03111 0.36500 0.31859 0.50722 1.00000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.9428 0.9928 0.9950 0.9944 0.9967 1.0000
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.02722 0.06111 0.07349 0.10000 0.84667
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.02889 0.06333 0.07567 0.10333 0.68333
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001263 0.0000000 0.5505556
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001169 0.0000000 0.0588889
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0003355 0.0000000 0.3711111
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0001083 0.0000000 0.2022222
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9378 0.9917 0.9939 0.9934 0.9956 1.0000
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01722 0.04389 0.06147 0.08111 0.84611
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01778 0.04556 0.06279 0.08444 0.71056
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 1.688e-05 0.000e+00 1.439e-01
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 4.741e-06 0.000e+00 7.778e-03
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 5.669e-06 0.000e+00 1.667e-03
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 8.84e-05 0.00e+00 1.00e+00

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9578 0.9922 0.9939 0.9938 0.9956 0.9994
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.03056 0.04681 0.06167 0.89167
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01333 0.03333 0.04830 0.06556 0.61778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 9.990e-07 0.000e+00 1.667e-02
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 1.549e-07 0.000e+00 1.667e-03
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.543e-06 0.000e+00 5.556e-03
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.020e-07 0.000e+00 4.444e-03
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5094 0.7000 0.7511 0.7423 0.7922 0.8822
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008333 0.022222 0.035896 0.045556 0.855556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

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##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.023333 0.036711 0.048889 0.502778
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0009271 0.0000000 0.7466667
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.3744 0.6094 0.5584 0.7722 1.0000
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0002462 0.0000000 0.1877778
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.001878 0.000000 1.000000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9683 0.9917 0.9933 0.9934 0.9950 0.9994
##
## beta.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007222 0.017222 0.030347 0.037222 0.857778
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.018889 0.031161 0.040000 0.487778
##   Modality 2 :
##     Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9739 0.9917 0.9933 0.9931 0.9950 0.9989
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.013889 0.025893 0.031667 0.745556
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.014444 0.026777 0.033333 0.445000
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 3.950e-06 0.000e+00 2.239e-01
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 9.106e-05 0.000e+00 1.428e-01
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 2.72e-07 0.00e+00 1.00e-02
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9556 0.9906 0.9928 0.9923 0.9944 0.9994
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00000 0.00500 0.01278 0.02645 0.02833 0.82167
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.005556 0.013333 0.026209 0.031111 0.648889
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.00e+00 0.00e+00 0.00e+00 3.53e-05 0.00e+00 1.00e+00
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      0      0      0      0      0      0
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0000247 0.0000000 0.5366667
##
## ‘‘‘

```

Conclusion

This brief report includes acceptance rate summaries, posterior probabilities, and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.